

CMM707

Academic Year	2026
Semester	3
Module Number	CMM707
Module Title	Cloud Computing
Assessment Method	Coursework
Deadline (time and date)	22/07/2026, 11:00 PM IST - For the report 25/07/2026, 09:30 AM onwards Viva will commence
Submission	Assessment Dropbox in the Module Study Area in CampusMoodle.
Word Limit (see Assessment Word Limit Statement)	Not more than 2000 words (Including images, references)
Use of Generative Artificial Intelligence (Gen AI) See Student Guidance on the use of AI	1 2 3 4 5 No AI Ideas Editing Tasks Full AI ☐ ☐ ☐ ☐ ☒
Module Co-ordinator	Lasitha Petthawadu (Module Leader) & Semini Perera

What knowledge and/or skills will I develop by undertaking the assessment?

Fundamental theoretical and practical knowledge on Cloud computing starting from the foundational aspects of the subject area.

- You would be able to differentiate different approaches to migrating and deploying applications on the cloud.
- Understand compute related services both server based and serverless such as AWS EC2, AWS Lambda.
- Understand data related services ranging from Object storage to block storage services such as AWS S3, AWS EBS.
- Understanding of databases and DBMS solutions available on the cloud such as relational, Non relational.
- Understanding of Cloud Design Patterns and Principles.
- Practically understand the implementation of non functional aspects such as scalability, availability on cloud environments using load balancers, auto scaling groups.
- Introduction to cloud containerization and working with Docker.
- Detailed introduction to container orchestration on Kubernetes.
- Introduction to Microservices & the difference between monolithic architectures vs microservices.
- Understanding and implementation of containerization in microservices architecture

On successful completion of the assessment students will be able to achieve the following Learning Outcomes:

1. Compare the operational concepts, implementation and performance issues of cloud computing systems, and the relative merits and suitability of each for complex data-intensive applications.
2. Critically appraise different cloud computing models, namely, infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS)
3. Evaluate design choices when solving real-world cloud computing problems by analysing and contrasting different cloud computing solutions.
4. Integrate software components in novel ways to architect and develop cloud-based applications solutions for an enterprise
5. Security and ethics of cloud computing with public cloud solutions, Legal implications of off-premise hosted data, impact on society implications and benefits

Please also refer to the Module Descriptor, available from the module Moodle study area.

What is expected of me in this assessment?

Task(s) - content

This coursework aims to validate the theoretical and the practical aspects discussed during the labs and the lectures of this module.

Case Study

New Event (<https://templatemo.com/tm-486-new-event>) is a responsive, single-page HTML template designed for events, conferences, and workshops. It features sections including Watch Videos, Creative Speakers, Our Programs, and Register Here.

You have been engaged as a cloud engineer to support the deployment of this frontend application alongside a set of backend microservices and an analytics infrastructure.

You are tasked with designing a microservice architecture that will run on a Kubernetes cluster. Your solution should take into account the following attributes:

- Scalable
- Secure
- Fault-Tolerant
- Affordable

Solution Requirements:

1. Deployment of the Frontend

Your frontend application should be deployed within the Kubernetes cluster and served as a single endpoint.

2. Microservices Design

The architecture should be divided into multiple microservices, including but not limited to:

- **Event Service**
Stores and updates event details such as Event ID, Title, Venue, Date/Time, Ticket Price, Capacity, and Seats Available.
- **Program Service**
Provides agenda data for the "Our Programs" section including Day, Track, Session, Speaker Name, and Times. (Track means a category, for example: Cloud Computing Track)
- **Registration Service**
Saves attendee registrations including Registration ID, Event ID, Name, Email, Ticket Count, and Timestamp.

What is expected of me in this assessment?

When the **SeatsAvailable** count for an event drops below a set threshold (e.g. < 10 seats left), it should trigger a serverless function.

The function must perform one of the following actions:

- Send a simple notification email using a free-tier service such as AWS SES or SendGrid
- OR**
- Write a notification entry into cloud object storage (e.g., an S3 bucket) with the event ID, timestamp, and remaining seats.

3. Saving Transactional Data

Your Event Service, Program Service and Registration Service must save data on a Relational Database (any cloud-hosted relational DB of your choice).

It is not mandatory to integrate these services with the frontend web app.

4. Capturing Web Analytics

You have to capture at least three types of web analytics from the frontend and stream it to ClickHouse (open-source and free to self-host). The web analytics that you capture will be evaluated based on how useful and relevant it is for the use case. Make sure to differentiate your metrics from others in your cohort.

5. Visualization of Analytics

Data Visualization: You can use one of the following: Metabase, Apache Superset, AWS Quick Sight or Power BI to create real-time, interactive dashboards for visualizing the data captured through ClickHouse. The dashboards should provide meaningful insights for the given scenario.

What is expected of me in this assessment?

6. Observability

Deploy observability infrastructure within the Kubernetes cluster to monitor the health, performance, and availability of all services.

7. CI/CD Pipeline

This solution requires an automated service build and deployment using a CI/CD pipeline, maintaining 100% uptime using a Blue-Green deployment strategy.

Task(s) - format

Students are tasked to write a report to describe the designed solution.

1. The report needs to be in the format of a PDF file.

What is expected of me in this assessment?

2. Students should include the solution architecture diagram and the deployment architecture diagram into the report. In the solution diagram the student has to highlight the request flows and the data flows. [LO1]
3. Include a discussion on security and ethics challenges that might arise during the implementation of the New Events platform. [LO1]
4. Include the CI/CD pipeline designs diagrams and CI/CD process descriptions into the same report under CI/CD process section. Highlight security and ethics challenges of this cloud solution. [LO1][LO2][LO3][LSEPI]
5. Implement the services and frontend and create Kubernetes artefacts required to deploy the designed solution into a Kubernetes cluster. [LO4]
6. Implement a CI/CD pipeline to deploy the created solution. This pipeline should be able to deploy the artefacts to the production environment following the Blue-Green model. [LO4]
7. The report should consist of scripts used to deploy the CI/CD pipeline. [LO4]
8. Create a simple runbook with the steps to deploy the designed solution and test it. Students have to include this runbook in the final report. [LO4]
9. Evidence of ClickHouse deployed, and web analytics captured.
10. Evidence of the visual analytics dashboard built using any of the above suggested platforms that visualizes the analytics data captured on ClickHouse.

You are not allowed to demonstrate your solution locally on any local setup and need to utilize a cloud cluster. At the time of the viva, you should have a demonstrable solution on any Cloud Platform of choice (eg: AWS, MS Azure, etc.)

Minikube is used during labs for demonstration purposes but for this assignment you're required to use a scalable Kubernetes setup (cloud native - EKS, GKS, AKS, microk8s, k3s).

AI tools may be used to assist your work. However, you must be able to explain and justify every aspect of your solution. If it is evident during the viva that any part of the solution is AI-generated and you cannot demonstrate understanding of it, marks will be deducted accordingly.

Vivas are mandatory, failing to attend the viva session would result in the coursework being graded as F.

How will I be graded?

A grade will be provided for each criterion on the feedback grid which is specific to the assessment. The overall grade for the assessment will be calculated using the algorithm below.

How will I be graded?	
A	At least 75% of the feedback grid to be at Grade A, at least 100% of the feedback grid to be at Grade B or better
B	At least 75% of the feedback grid to be at Grade B or better, at least 100% of the feedback grid to be at Grade C or better
C	At least 75% of the feedback grid to be at Grade C or better, and at least 100% of the feedback grid to be at Grade D or better.
D	At least 75% of the feedback grid to be at Grade D or better, and at least 100% of the feedback grid to be at Grade E or better
E	At least 100% of the feedback grid to be at Grade E or better.
F	Failing to achieve at least 100% of the feedback grid to be at Grade E or better or not attending the scheduled viva session.
NS	Non-submission.

Feedback grid

GRADE	A	B	C	D	E	F
DEFINITION / CRITERIA (WEIGHTING)	EXCELLENT Outstanding Performance	COMMENDABLE/VERY GOOD Meritorious Performance	GOOD Highly Competent Performance	SATISFACTORY Competent Performance	BORDERLINE FAIL	UNSATISFACTORY Fail
Participation for the viva Weight 1	Student has participated on time, is ready with the demonstration.	Student participated for the viva on time but took time to start his demonstration due to hardware/software, online screen sharing issues.	Student has participated for the viva late and was ready with the demonstration.	Student was late for the viva and was not ready to demonstrate and took time after being late as well.	Student rescheduled the viva at the last moment without prior notice. and joined late on the rescheduled date or had trouble starting the demonstration on time	Student did not participate to the viva on the scheduled date
Identifying Web Analytics Weight 1	The Student has managed to identify 3 or more than 3 analytics that are relevant to the coursework expectation, and justifies them accurately.	Student has managed to identify 3 or more than 3 analytics and manages or attempts to give a reasonable justification	The student has identified 3 analytics which are relevant to the coursework expectations but fails to justify the relevance of them to the coursework expectation	The student has identified 3 analytics out of which at least one metric is not relevant to the coursework expectations and has failed to justify	The Student has identified less than 3 analytics.	The student has failed to identify any analytics.
Report Outline Weight 2	The student has clearly organised and outlined the report, meeting all the delivery criteria	The overall report outline is achieved but the student has made minor errors in the main sections of the report.	The Student has missed 25% of the main outcomes outlined as	The Student has missed 50% of the main outcomes outlined as	The Student has missed 75% of the main outcomes outlined as	The Student has missed 90% of the required content assessed in the report.

GRADE	A	B	C	D	E	F
DEFINITION / CRITERIA (WEIGHTING)	EXCELLENT Outstanding Performance	COMMENDABLE/VERY GOOD Meritorious Performance	GOOD Highly Competent Performance	SATISFACTORY Competent Performance	BORDERLINE FAIL	UNSATISFACTORY Fail
	outlined in the assessment content. Adhering to the word count	Not adhering to the word count	expectations of the report.	expectations of the report.	expectations of the report.	
Solution Implementation Weight 3	The solution is detailed and the student has covered a broader depth including functional as well as non-functional aspects	The solution implemented covers what is expected as well as tried to cover a broader depth with minor errors that have been highlighted during the viva	The solution implemented covers what is expected but the student has made major errors in one area of the assignment due to lack of understanding or misinterpretation	The solution implemented covers the bare minimum that is expected by the scope of this coursework but student fails to denote understanding of the implementation	The solution is partially functional with a majority of the functional and non-functional requirements not been met	The solution is not functional or is not a workable solution
Demonstration Weight 1	The student is able to demonstrate in a clear, understandable manner following an organised logical presentation sequence. The student was able to intuitively answer all the questions raised during the viva	The student is able to demonstrate clearly and was able to completely demonstrate all the requirements. But the student made minor errors during the viva demonstration.	The student was able to demonstrate clearly and was able to completely demonstrate all the requirements but had major	The student demonstrates more than 50% of the functionality but fails in answering questions accurately.	The student failed to demonstrate 50% of the expected functionality.	The student failed to demonstrate more than 75% of the functionality

Coursework received late, without valid reason, will be regarded as a non-submission (NS) and one of your assessment opportunities will be lost.



What else is important to my assessment?

What is plagiarism?

"Plagiarism is the practice of presenting the thoughts, writings or other output of another or others as original, without acknowledgement of their source(s) at the point of their use in the student's work. All materials including text, data, diagrams or other illustrations used to support a piece of work, whether from a printed publication or from electronic media, should be appropriately identified and referenced and should not normally be copied directly unless as an acknowledged quotation. Text, opinions or ideas translated into the words of the individual student should in all cases acknowledge the original source" ([RGU 2022](#)).

What is collusion?

"Collusion is defined as two or more people working together with the intention of deceiving another. Within the academic environment this can occur when students work with others on an assignment, or part of an assignment, that is intended to be completed separately" ([RGU 2022](#)).

For further information please see [Academic Integrity](#).

What is the Assessment Word Limit Statement?

It is important that you adhere to the Word Limit specified above. The Assessment Word Limit Statement lists what is included and excluded from the word count, along with the penalty for exceeding the upper limit.

What if I'm unable to submit?

- The University operates a [Fit to Sit Policy](#) which means that if you undertake an assessment then you are declaring yourself well enough to do so.
- If you require an extension, you should complete and submit a [Coursework Extension Form](#). This form is available on the RGU [Student and Applicant Forms](#) page.
- Further support is available from your Course Leader.

What additional support is available?

- [RGU Study Skills](#) provide advice and guidance on academic writing, study skills, maths and statistics and basic IT.
- [RGU Library guidance on referencing and citing](#).
- [The Inclusion Centre: Disability & Dyslexia](#).
- Your Module Coordinator, Course Leader and designated Personal Tutor can also provide support.

What are the University rules on assessment?

The University Regulation '[A4: Assessment and Recommendations of Assessment Boards](#)' sets out important information about assessment and how it is conducted across the University.